

Sam Son

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RESEARCH INTERESTS

I am interested in designing and analyzing systems that improve the cost efficiency of modern CPU/GPU infrastructure, with a current focus on large-scale ML inference. My recent work approaches this problem by analyzing the tradeoff in performance and cost of GPU interconnects and developing co-location techniques for large-scale MoE inference. Previously, I worked on fine-grained CXL memory management and software techniques for harvesting CPU memory-stall cycles. Keywords: ML Systems, LLM Inference, Operating Systems, Resource Management, Cost-Efficient System Design.

EDUCATION

University of California, Berkeley Ph.D. in Electrical Engineering and Computer Sciences Advisors: Scott Shenker and Sylvia Ratnasamy	Berkeley, CA <i>Aug 2022 – Expected Dec 2026</i>
Seoul National University M.S. in Computer Science & Engineering Advisor: Jae W. Lee	<i>Mar 2020 – Aug 2022</i>
POSTECH B.S. in Convergence IT Engineering, <i>Summa Cum Laude</i>	<i>Mar 2014 – Feb 2020</i>

RESEARCH & INDUSTRY EXPERIENCE

University of California, Berkeley <i>Graduate Student Researcher</i>	Berkeley, CA <i>Aug 2022 – Present</i>
• Current project: Developing GPU-sharing techniques for large-scale LLM inference, with a focus on improving GPU utilization by co-locating decode-dominant MoE inference with complementary inference workloads. The key challenge is the parallelism mismatch between highly expert-parallel MoE inference and smaller co-located workloads. • Prior research: Built systems and performance models spanning MoE inference networking, CXL memory management, and CPU resource harvesting.	
Meta Platforms, Inc. <i>Part-time Student Researcher, Meta Superintelligence Lab</i>	Remote <i>Aug 2025 – Dec 2025</i>
• Proposed and evaluated spatial GPU sharing for improving resource efficiency in large-scale generative AI inference.	
Meta Platforms, Inc. <i>Research Scientist Intern, AI-Systems Co-design</i>	Menlo Park, CA <i>May 2025 – Aug 2025</i>
• Contributed to developing high-fidelity performance model for LLM inference; Evaluated the benefit of scale-up GPU systems such as NVL72 using this framework.	
Meta Platforms, Inc. <i>Research Scientist Intern, AI-Systems Co-design</i>	Menlo Park, CA <i>May 2024 – Aug 2024</i>
• Contributed to the design of a cacheline-granularity CXL memory-management system.	

SELECTED PUBLICATIONS

Rethinking Network Topologies for Cost-Effective Mixture-of-Experts LLM Serving | *arXiv preprint, 2026*
• J. Choi[†], **S. Son**[†], S. Choi, H. Kim, Y. S. Shao, S. Shenker, S. Ratnasamy, B. Nikolic. [arXiv:2605.00254](#)
• Evaluated the performance and cost tradeoffs of different GPU network topologies in MoE LLM inference using the performance model.

Extending Managed Runtimes for Object-Level CXL Memory Management | *EuroSys 2027 (to appear)*
• **S. Son**, Z. Luo, W. Zhang, S. Ratnasamy, S. Shenker.
• Designed and implemented object-level CXL memory management for managed runtimes, improving the cost-effectiveness of CXL tiered memory systems.

Harvesting Memory-bound CPU Stall Cycles in Software with MSH | *OSDI 2024*
• Z. Luo, **S. Son**, S. Ratnasamy, S. Shenker.
• Developed a software hyperthreading system that dynamically harvests memory-stall cycles without the rigidity of hardware hyperthreading.

Efficient Microsecond-scale Blind Scheduling with Tiny Quanta | *ASPLOS 2024*
• Z. Luo, **S. Son**, D. Bali, E. Amaro, A. Ousterhout, S. Ratnasamy, S. Shenker.
• Developed a CPU scheduler for microsecond-scale workloads that minimizes scheduling overhead using compiler-assisted cooperative preemption.

FlashNeuron: SSD-Enabled Large-Batch Training of Very Deep Neural Networks | *FAST 2021*
• J. Bae, J. Lee, Y. Jin, **S. Son**, S. Kim, H. Jang, T. J. Ham, J. W. Lee.
• Developed an SSD-based tensor-offloading system that reduces GPU memory pressure, enabling larger training batches and higher DNN training throughput.

[†] Equal contribution. See Google Scholar for the complete publication list.

SELECTED HONORS & AWARDS

Korean Foundation for Advanced Studies (KFAS) Scholarship	2022 – 2027
Best Master's Thesis Award , Seoul National University CSE	Aug 2022
Best TA Award , College of Engineering, Seoul National University	Jun 2020

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Go, Shell
ML/GPU Systems: PyTorch, CUDA, vLLM, NCCL, Nsight Systems/Compute
Systems: Linux Kernel, GDB, ftrace, perf